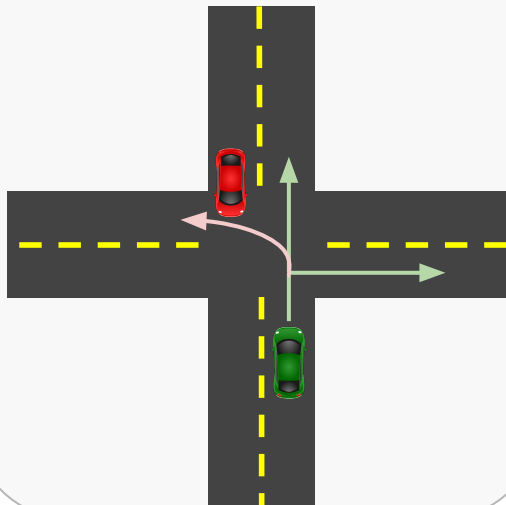
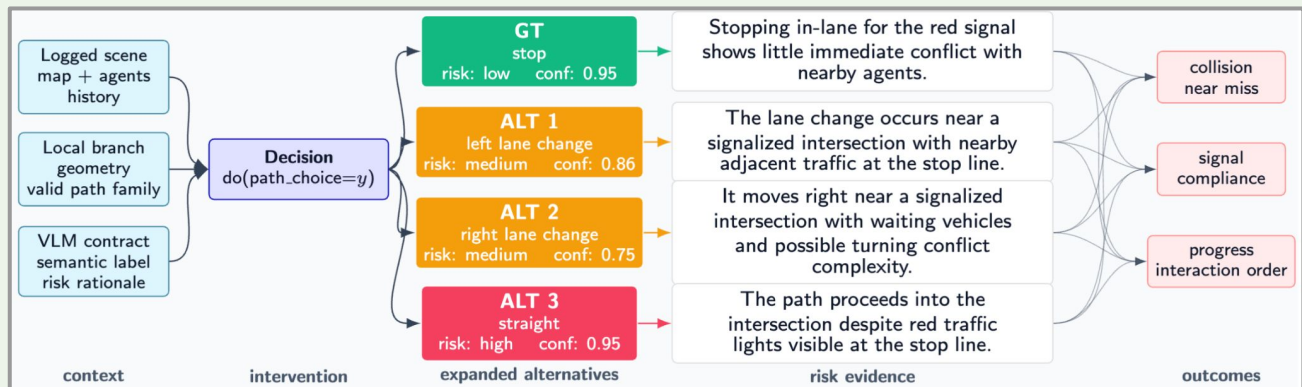


DAG-Based Causal Understanding

Unlabeled Road Geometry



FRONTIER VLM EXTRACTS MANEUVERS



Sampled Intervention Conditions Transformer

ALT 1
left lane change
risk: medium conf: 0.86

Semantic request
 $u = (y, W, m)$

Semantic embedding
 $z_y = E_{\text{sem}}(y)$

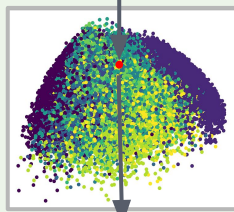
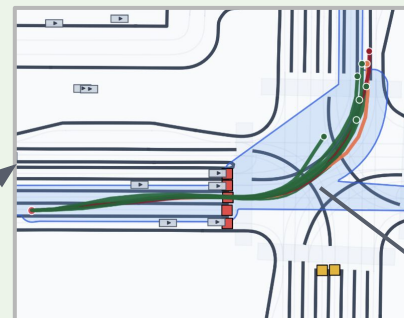
Mask-gated adapter
attention + bias
on $\Omega = W \times m$

Logged scene
 x

Scene encoder
 $c = f_{\text{enc}}(x)$

Autoregressive
decoder

Trajectory
rollout τ



Advantages